

Mix It Up. On Purpose.

Interleaving · Kornell & Bjork · Rohrer & Taylor

"I studied Floating pulses for an hour
and aced the practice problems.
Then the quiz asked me to tell Floating from Superficial."

— Blocked practice builds fluency within a category. Interleaving builds discrimination between them.

The most natural way to study is to finish one topic before moving to the next — all Depth pulses, then all Rate pulses, then all Shape pulses. This is called **blocked practice**. It feels efficient. It produces fast short-term gains. And in the research, it consistently underperforms **interleaved practice** — where you mix categories deliberately — on every test that matters: delayed recall, transfer, and discrimination between similar concepts. Exactly the conditions of a real quiz.

The Research: Why Mixing Beats Blocking

Kornell and Bjork (2008) had participants learn to identify paintings by artist. Blocked learners studied all paintings by each artist in sequence. Interleaved learners saw paintings mixed randomly. Blocked learners performed better during practice. Interleaved learners performed dramatically better on the final test — and most of them mistakenly believed the blocked approach had worked better for them.

Rohrer and Taylor (2007) replicated this in mathematics: students who practiced problems in mixed order performed 43% better on a delayed test than students who practiced in blocked order, despite equivalent study time.

+43%

better delayed test performance with
interleaving vs. blocked practice (Rohrer
& Taylor, 2007)

feels ↓

Interleaving feels harder and less
productive while studying — which is
exactly why most students avoid it

discrimin.

Interleaving teaches your brain WHICH
strategy to use, not just how to execute it
— the key clinical skill

Why It Works: The Discrimination Problem

When you study blocked — all Depth pulses together — your brain doesn't need to identify **which category** the current item belongs to. The category is given. On a quiz (or with a real patient), the category is not given. You have to figure out whether what you're feeling is Floating, Superficial, or something else entirely before you can apply what you know about it.

Interleaving forces your brain to solve this categorization problem on every single item during practice. After enough mixed practice, the discrimination becomes automatic — you feel something under your finger and your brain starts sorting it into the right category before you consciously decide to. That's the clinical skill you actually need.

The "Which one is this?" effect: Blocked practice answers "How do I handle a Floating pulse?" Interleaved practice answers "Is this a Floating pulse or something else — and how do I tell?" The second question is what the quiz asks. It's what patients require.

Applied to the 28 Pulse Qualities

Blocked (less effective)	Interleaved (more effective)
Study all Depth pulses: Floating, Superficial, Leather, Deep, Firm, Hidden	Mix: Floating → Knotted → Slippery → Deep → Wiry → Hasty → Flooding → Tight
Practice 10 Floating pulse questions, then 10 Deep pulse questions	Practice quiz randomizes all 28 qualities — you have to identify the category before applying knowledge
Learn Knotted / Hasty / Intermittent as a group, then move on	Return to Knotted 2 days later, mixed with other qualities — forces re- discrimination from scratch

The practice quiz is interleaving built in. A randomized 25-question quiz on all 28 qualities — where you don't know what's coming next — is one of the highest-yield study activities available for this material. The discomfort of not knowing what category is coming is the mechanism, not a side effect.

The confidence trap (again): Blocked practice feels better because you're comparing items within a category where the differences are obvious. Interleaved practice feels harder because you're comparing across categories where the differences are subtle. The harder feeling is the one that produces the A+.

How to Interleave Your Pulse Study

Flashcard shuffle: Make one card per quality. Shuffle completely before every session. Never study them in the same order twice. If you find yourself predicting what comes next, the deck isn't shuffled enough.

Practice quiz randomization: The course practice quiz randomizes questions by design. When it feels harder than studying your notes — that's not a design flaw, that's the feature.

Journal interleaving: When taking pulses for journal entries, consciously compare the current finding to a different quality type each time. "This feels Slippery — but how do I know it isn't Flooding? What would Flooding feel like here?"

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